

Stat331 Assignment 2

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Set up

Here are the libraries used:

```
library(data.table)
```

```
##
## Attaching package: 'data.table'

## The following object is masked from 'package:base':
##
##      %notin%
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::between()      masks data.table::between()
## x dplyr::filter()       masks stats::filter()
## x dplyr::first()        masks data.table::first()
## x lubridate::hour()     masks data.table::hour()
## x lubridate::isoweek()  masks data.table::isoweek()
## x lubridate::isoyear()  masks data.table::isoyear()
## x dplyr::lag()          masks stats::lag()
## x dplyr::last()         masks data.table::last()
## x lubridate::mday()     masks data.table::mday()
## x lubridate::minute()   masks data.table::minute()
## x lubridate::month()    masks data.table::month()
## x lubridate::quarter()  masks data.table::quarter()
## x lubridate::second()   masks data.table::second()
## x purrr::transpose()    masks data.table::transpose()
## x lubridate::wday()     masks data.table::wday()
## x lubridate::week()     masks data.table::week()
## x lubridate::yday()     masks data.table::yday()
## x lubridate::year()     masks data.table::year()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Here is also my A1 Question 2 code copy-pasted in, so that I can complete A2 Question 1 without hassle:

```

#2.FROM A1
truebeta <- c(-200000, 45, -300, 10000, 150000)
truesigma <- 12000
A1_variates <- read_table('~Documents/GitHub/Big-Ahh-R-Repo/Stat331/Stat331 A1/data/raw/A1_variates (1

##
## -- Column specification -----
## cols(
##   size = col_double(),
##   age = col_double(),
##   employees = col_double(),
##   col = col_double()
## )

n <- nrow(A1_variates)
X <- as.matrix(cbind(rep(1,n),A1_variates))
set.seed(21066922)

#a)
simulatedresponse <- (as.vector(X %*% truebeta) + rnorm(n,0,truesigma))
#this is the simulated data with the random seed set to my student number
simulatedresponse

## [1] 166281.206 153730.937 43338.698 193046.302 137185.704 10542.334
## [7] 14183.044 81275.444 115711.040 98465.901 39357.221 11103.992
## [13] 7704.852 82777.187 110100.768 88648.310 47059.937 161115.200
## [19] 160731.747 142684.940 123764.878 126090.364 173399.992 -5527.173

#b)
y <- simulatedresponse
size <- A1_variates$size
age <- A1_variates$age
employees <- A1_variates$employees
col <- A1_variates$col
mlr <- lm(y~size+age+employees+col)
summary(mlr)

##
## Call:
## lm(formula = y ~ size + age + employees + col)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22388  -5899   1651   7308  17474
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -161990.95   58235.05  -2.782 0.011889 *
## size          66.05      17.06    3.872 0.001027 **
## age        -447.16     392.25   -1.140 0.268453
## employees    7100.09    1657.67    4.283 0.000401 ***
## col         113817.43   54698.58    2.081 0.051221 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 11500 on 19 degrees of freedom
## Multiple R-squared: 0.9709, Adjusted R-squared: 0.9648
## F-statistic: 158.5 on 4 and 19 DF, p-value: 2.6e-14
```

```
#c)
betahat <- solve(t(X)%*%X)%*%t(X) %*% y
#d)
confint(mlr)
```

```
##              2.5 %      97.5 %
## (Intercept) -283878.3251 -40103.5843
## size         30.3454    101.7547
## age          -1268.1392    373.8184
## employees     3630.5454  10569.6272
## col          -668.0088  228302.8661
```

FROM HERE ON IS THE START OF A2

Question 1

Using the provided code, and my student number, I generated the betas:

```
betahat_mat = array(0,c(5,1000)) #creates an array to store sets of estimates
set.seed(21066922)
for (i in 1:1000){
  y<-X%*%truebeta+rnorm(24,0,12000) #X and Beta vector as defined in A1.
  A1sim.lm<-lm(y~size+age+employees+col)
  betahat_mat[,i]<-coef(A1sim.lm) #yields estimates for each iteration
}
```

a)

Then we will specifically focus on the beta_2:

```
betahat_mat[2,]

##      [1] 66.05002666 43.87575653 53.01144804 47.98280338 32.37587784
##      [6] 67.50546009 31.93719334 51.60804551 52.00534788 51.25325086
##     [11] 41.07436771 77.85328624 68.99281833 64.50584605 46.98548077
##     [16]  1.89186411 49.76301711 27.65918151 69.06511279 74.89605545
##     [21] 57.18067235  4.96857609 76.61670648 12.57508156 51.21112114
##     [26] 49.58377011 53.74398113 72.25653703 64.29875796 17.27724366
##     [31] 60.57388200 36.22716352 65.75930465 21.75794870 67.08999078
##     [36] 63.03989818 30.34586795 78.64890158 45.83499172 39.24445780
##     [41] 20.10589045 62.57085647 17.32880691 42.47797268 43.57534683
##     [46] 25.73766720 20.42872371 44.60273587 45.86064908 43.68680076
##     [51] 21.05761136 52.83377464 17.48645763 27.96584284 30.76073710
##     [56] 48.96715565 50.66304408 51.42739239 40.30615210 21.07661199
##     [61] 48.62424881 57.81855140 40.31752268 50.36861568 31.28243797
##     [66] 40.17018696 67.65682371 33.81556358 60.45862474 41.88867444
##     [71] 60.36755746 63.78715438  8.76971443 67.86159655 47.34652871
##     [76] 61.50764422 71.26067585 52.20619345 56.29331384 80.44137796
##     [81] 82.90692654 -11.81071311  9.64201490 62.19466985 17.23150646
##     [86] 42.41133281 45.50188253 47.43569632 53.58106775 79.34996789
##     [91] 43.57481518 55.29354634 32.02552815 69.93540881 33.45036034
```

##	[96]	39.18829818	54.01645704	62.94400583	54.89422546	49.87968386
##	[101]	42.50389674	37.46843179	36.68911391	38.89780087	41.49852386
##	[106]	28.57632921	81.56414562	56.21055545	76.14714679	19.14294637
##	[111]	56.42945941	29.45026582	65.96582992	68.38813814	55.23774650
##	[116]	28.70866345	48.67279126	54.61572798	14.72281161	13.96459655
##	[121]	41.17218071	45.32011407	61.34798420	31.73759768	47.33506661
##	[126]	55.04185132	56.99061249	21.17313407	35.16774482	41.54268378
##	[131]	70.57061401	39.28327712	34.85183717	41.99877195	58.31163626
##	[136]	34.48157121	73.13136281	65.14549975	42.97390691	44.62832871
##	[141]	31.49855008	51.50321694	36.87758818	70.22924187	33.16676867
##	[146]	34.13119545	67.25877165	18.69807303	40.74037853	31.13743232
##	[151]	43.95332102	80.15870739	48.62775275	47.85792477	44.19382390
##	[156]	90.11922894	37.01952826	36.10992222	73.35913699	23.04266279
##	[161]	47.59194860	29.60900135	41.63545573	51.39709750	54.21659885
##	[166]	23.68104078	36.78007851	9.42897183	57.60124873	36.02331405
##	[171]	68.27324155	59.94623595	67.66564338	71.07045647	41.76413273
##	[176]	18.10925479	51.66284030	23.22870833	66.35404610	27.89588618
##	[181]	0.02244654	56.13507557	59.32509369	49.15844083	34.47738991
##	[186]	70.77713858	60.87499598	45.40166940	56.61639815	69.62940355
##	[191]	43.39423313	42.22138826	30.41062344	40.95867176	34.29407378
##	[196]	67.97886217	31.10773191	49.24739898	37.27537607	68.10759464
##	[201]	55.15410448	63.10062544	32.29868082	41.74946778	47.43396011
##	[206]	19.70990912	61.27382978	61.90409030	53.84990391	46.10773959
##	[211]	48.55712469	38.13305864	56.62369259	38.63963362	31.81839733
##	[216]	45.45138470	74.73290412	54.05456602	45.14022077	48.38973087
##	[221]	36.66643298	21.15733337	55.12573015	38.68445757	35.70540581
##	[226]	51.87692603	29.89100728	65.72946252	13.71567680	26.03017329
##	[231]	38.50992445	64.53482773	9.02655558	9.73809244	50.16642576
##	[236]	62.73271551	76.73053230	72.95957613	30.78923116	72.85367844
##	[241]	10.78099853	32.61145940	45.42009849	48.48555436	56.00081048
##	[246]	45.90467963	66.21795585	53.45384203	25.62151895	42.34443018
##	[251]	12.83012285	28.10840729	96.66725066	41.36470817	56.85206060
##	[256]	71.28829670	42.35773024	62.11790832	55.55955463	68.51742012
##	[261]	16.45881228	85.52128306	57.26889316	20.19008359	37.62104702
##	[266]	65.40005378	38.28277851	48.14135794	65.91406645	60.35034073
##	[271]	62.44469169	50.20705201	65.73631191	5.28213203	28.10792228
##	[276]	44.51265128	45.46833325	31.15502558	78.07515510	44.71416098
##	[281]	61.32781803	42.25563712	60.77871662	55.82124424	29.43281698
##	[286]	59.93937112	55.61674621	57.76930863	25.45914105	41.44500228
##	[291]	24.48909966	21.03955246	62.20011271	78.47422814	94.30739221
##	[296]	37.04963955	16.48939812	56.59708361	59.64647162	45.76835962
##	[301]	12.46895636	34.46644153	60.12718728	56.53920133	37.28462841
##	[306]	67.00257869	42.38554544	38.16058920	62.61278268	48.70985422
##	[311]	29.26701993	45.65926732	24.13657523	19.59379265	68.98526885
##	[316]	34.23828156	50.15645998	36.67236078	27.29212216	46.02895545
##	[321]	62.37660527	60.63912369	59.63476171	53.75366168	58.92549370
##	[326]	73.95522230	45.95820675	30.87056271	41.64469990	62.06978528
##	[331]	32.97965041	39.33180076	47.77426151	45.32920221	13.78919433
##	[336]	52.39967380	27.93337494	-0.80018005	63.46755128	51.80356951
##	[341]	32.18910076	48.26895591	10.79588951	83.14692259	15.51640053
##	[346]	32.08699796	52.40106445	6.57578349	41.66478443	60.38975371
##	[351]	50.49285614	47.31931991	10.84500360	54.69730957	41.24491268
##	[356]	36.59289492	50.33558785	61.16576298	53.03582337	31.03262455
##	[361]	43.81204201	70.50876824	29.09237705	44.75612082	48.96354496

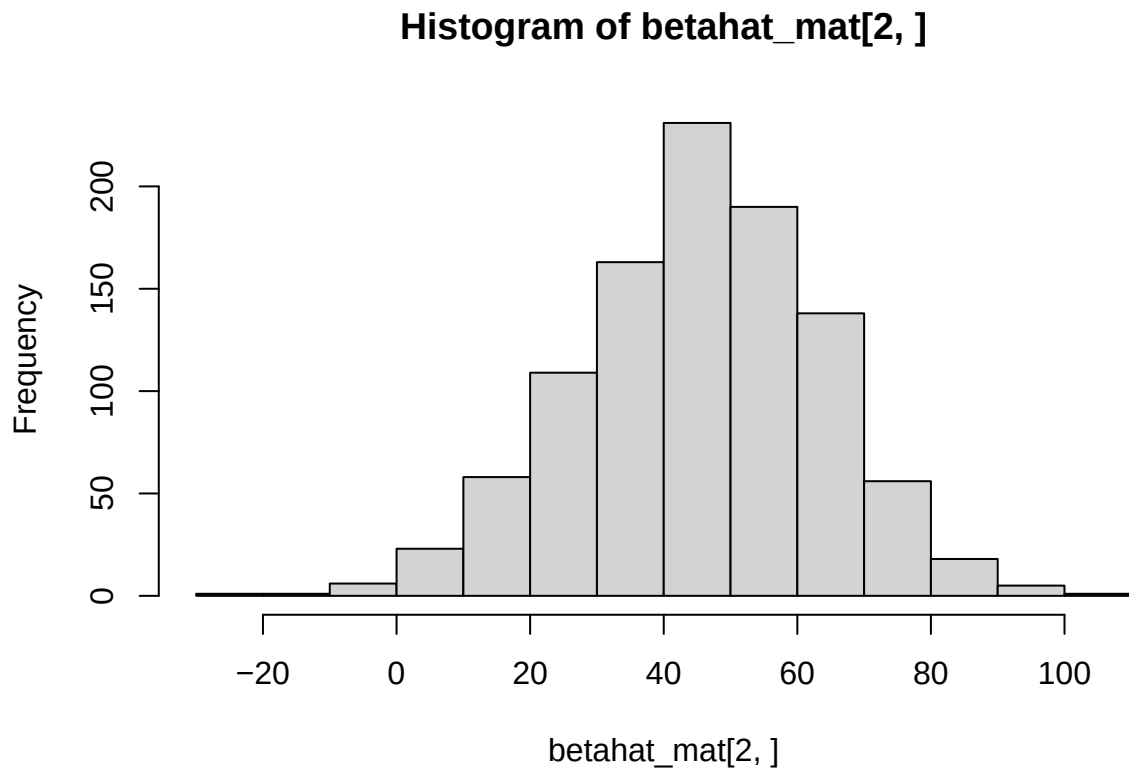
##	[366]	62.38652535	20.65106858	41.93567979	64.07137343	14.64421224
##	[371]	53.14571367	52.77567289	36.08726853	31.18345517	48.10564634
##	[376]	46.88556349	69.77720733	42.39051552	26.11134892	45.60992265
##	[381]	39.81141866	36.93657939	38.96878767	33.22539465	25.25014290
##	[386]	33.55681756	58.64319082	-1.94193738	35.43194561	53.27258443
##	[391]	55.96865689	64.08956530	29.08951008	46.35887115	39.50814481
##	[396]	66.76667828	72.75540956	-0.34787953	48.20304003	17.64555141
##	[401]	52.40902318	45.32920091	49.41275571	30.99238614	50.12092642
##	[406]	41.20488685	49.86891927	40.94258508	63.67700862	48.65020337
##	[411]	40.73850466	2.59069120	41.82222372	47.62328342	50.06931019
##	[416]	31.02221237	36.26525674	37.16364840	44.51867756	8.26351328
##	[421]	68.09373959	51.30197077	21.18518664	28.09210251	32.85519025
##	[426]	35.30242889	13.58376893	49.92759725	27.24948337	25.92573561
##	[431]	65.14951859	50.98748691	45.27189483	67.75975318	49.12341849
##	[436]	32.93910521	46.68729247	44.52572800	40.11555341	19.68709553
##	[441]	32.39513039	54.39516697	38.45382590	65.32290604	46.61071279
##	[446]	28.51903434	52.21496076	4.29206939	55.79754527	30.33572985
##	[451]	61.34178498	43.68514035	26.14440425	59.44565564	70.78596555
##	[456]	33.11825296	53.10081327	16.16144655	58.68943623	56.89369179
##	[461]	72.43682762	45.71121744	60.60666892	20.23609281	14.12553091
##	[466]	32.12966334	35.31764321	48.36205966	45.70533226	22.37485499
##	[471]	46.31609414	26.41663155	47.20818698	58.61024847	51.33819730
##	[476]	21.22421309	24.63601514	45.04999601	21.93717202	54.26448975
##	[481]	42.36188361	43.91382577	60.17054607	52.90724347	76.96165566
##	[486]	40.12528365	64.88868924	56.39918616	64.33186620	49.13270342
##	[491]	15.33040334	48.63619082	-4.57109694	90.90600479	48.27069492
##	[496]	15.38234734	46.07293588	80.63481897	42.43592390	63.22742584
##	[501]	48.36126805	38.10143596	73.90920780	18.45041955	57.01437167
##	[506]	63.24060326	51.87978562	64.58465240	80.01427661	55.55608280
##	[511]	37.45652775	28.41814270	34.61259311	50.45572871	47.89919068
##	[516]	49.53837056	61.15028813	44.28611849	62.19817522	56.88095015
##	[521]	53.61165730	83.82820656	40.63585101	19.50749141	8.71194298
##	[526]	88.89387804	53.32866051	17.24329198	20.01837824	52.26496053
##	[531]	17.22533565	56.34619174	75.78394803	48.68952649	66.76150196
##	[536]	36.81793483	35.35483403	17.58656725	29.54347290	72.85849036
##	[541]	47.73981131	26.02839075	55.70236348	30.02477371	52.45456343
##	[546]	73.70766235	55.14325136	20.60013872	37.15552183	36.84186123
##	[551]	17.51637226	14.72034397	30.11128292	52.04196131	28.92104149
##	[556]	36.56114657	47.74146687	51.96504555	51.32086295	43.79361740
##	[561]	30.90192678	22.73708934	48.94841432	30.35801213	51.37781823
##	[566]	48.11604352	31.84999727	62.11386711	21.84317782	54.53050252
##	[571]	52.85764413	47.80404207	64.56433356	31.79103705	11.12077985
##	[576]	36.35919231	53.81653997	4.75433911	47.69890168	74.65668805
##	[581]	11.69765169	8.43908221	59.57298789	65.96982822	64.93449753
##	[586]	41.08815425	37.21397629	42.86064252	2.52446767	29.70711755
##	[591]	32.45719721	57.49197326	63.45255516	42.53355354	71.19041723
##	[596]	51.66725983	57.13278899	54.80259946	65.93767043	54.90646576
##	[601]	58.78282250	63.70074387	47.31574410	67.00746866	79.42523663
##	[606]	40.43349372	41.98505013	66.93325466	45.21562792	36.01411022
##	[611]	64.09736168	12.66437923	17.69622454	50.14480901	37.60224457
##	[616]	56.77067640	43.67897438	56.28239367	29.15569634	51.00484898
##	[621]	40.18626706	49.25496025	58.87761051	38.41289923	36.49253359
##	[626]	64.77165621	66.41648574	54.15283515	44.55837645	37.89272157
##	[631]	33.04628337	21.79889150	55.24783704	80.74880692	26.42216764

##	[636]	41.98401104	25.64597030	46.99795980	50.59216855	65.26565984
##	[641]	49.28867776	71.58631342	46.98265214	52.59882935	30.39186240
##	[646]	64.73689712	46.01874338	26.89911044	89.75590244	75.53283512
##	[651]	28.12701490	28.59685662	17.83758500	56.34307791	39.92687689
##	[656]	34.52490826	57.75468953	29.53628331	51.65499360	59.44112482
##	[661]	10.22041579	24.04410049	87.73752102	53.90385410	47.98446530
##	[666]	60.25368569	26.84092681	47.88740113	29.54254693	38.69998851
##	[671]	61.76164522	49.13972770	55.25997349	62.66038752	40.58058996
##	[676]	46.03930783	64.01639300	55.92851501	61.69775613	41.00026551
##	[681]	41.03453326	-20.01430018	29.29461735	46.03279011	53.56688153
##	[686]	61.09205595	29.93310169	65.50132624	26.90921270	25.98964099
##	[691]	40.74978127	71.58883536	70.24503268	30.99750446	58.21333555
##	[696]	36.84013839	50.12395396	81.07896718	33.81174000	66.96246606
##	[701]	56.76372029	39.75498617	67.82983117	42.60757946	64.72530790
##	[706]	57.97127671	49.27462605	30.94190125	57.88042419	12.91830571
##	[711]	13.35693493	41.22600120	56.10665444	36.86275427	11.46446260
##	[716]	37.15440618	34.21062660	72.50310620	55.90815479	45.72296074
##	[721]	47.97652848	42.17227803	34.32460024	21.20481506	36.31046771
##	[726]	55.55208646	71.20873705	37.63272674	31.46041911	19.59670266
##	[731]	63.26608859	80.19209866	41.91674082	30.95363509	50.00519609
##	[736]	7.47979009	28.05640677	13.31318536	48.79004761	53.31773381
##	[741]	34.30772564	25.30396234	20.68984629	58.67121283	66.38938276
##	[746]	35.07480511	24.45638917	73.58779270	73.15935598	43.92092711
##	[751]	52.21071867	38.98016133	53.13868161	42.69290028	35.35704175
##	[756]	25.24654978	54.55092587	52.19972234	43.68409144	56.18358736
##	[761]	47.85667764	65.04486603	26.60726327	40.03424248	61.04671056
##	[766]	26.76242390	49.44918017	49.65359150	28.43606584	30.93731590
##	[771]	36.16011152	44.22621419	42.73503358	23.32708961	63.44565967
##	[776]	66.70096021	44.46257746	62.88388667	92.43021029	63.20378673
##	[781]	38.93569786	5.10744501	31.61325893	63.98052696	55.57493832
##	[786]	46.88933363	26.70958445	33.75376373	44.13275343	57.00340291
##	[791]	69.83111440	63.76978316	58.68552994	62.56241629	41.39601410
##	[796]	60.91366200	78.45125250	10.21321086	38.48894165	34.28514425
##	[801]	21.00324063	44.09165975	36.73171701	64.31342224	56.90536566
##	[806]	68.55430069	81.18945078	43.60962083	46.40393544	35.61477901
##	[811]	17.99784277	2.33509360	40.22859167	48.37834788	49.87030612
##	[816]	17.48703553	45.31625333	48.61422465	33.62644739	43.76469772
##	[821]	72.46027810	16.12069788	58.56704368	41.03769340	62.69611535
##	[826]	61.90705930	56.82762460	42.90831340	42.71992443	37.63647398
##	[831]	28.36461009	20.61652530	39.47189596	44.01157945	55.16805183
##	[836]	75.64070998	38.32853623	28.40331116	45.66252499	61.50289468
##	[841]	26.43369046	41.42326106	54.47147465	25.32758698	25.17725932
##	[846]	44.27644457	63.53555437	44.30929156	49.14676378	67.24061977
##	[851]	60.26591369	38.20965693	7.85374473	54.83647875	40.74110440
##	[856]	49.20167735	54.69988882	44.15805019	-3.16891483	41.83917151
##	[861]	46.96278252	63.35747336	52.07160551	58.70621084	41.09552959
##	[866]	70.27069875	56.92587972	58.81276029	-7.07910245	56.09196605
##	[871]	78.67373172	63.38816771	50.91204017	60.33851781	62.84316088
##	[876]	49.02506450	42.47715283	35.55971300	22.57355328	35.99450810
##	[881]	30.95765456	26.73825752	72.95984207	66.86079497	25.24275314
##	[886]	15.05056718	69.03393488	55.57888408	28.22892672	32.25103710
##	[891]	50.28533182	75.38179471	30.17083182	51.09983254	42.37760490
##	[896]	52.03796194	39.40973551	47.06205148	55.05515291	29.76115638
##	[901]	30.67889664	81.83957699	76.98777751	25.94958034	49.13710618

```
## [906] 42.62476870 79.57913498 34.57604451 72.06900037 39.67976909
## [911] 103.66459201 38.06336436 34.40993755 62.35701024 33.28320158
## [916] 28.48982387 58.78977829 37.72324971 73.14152165 47.08768253
## [921] 12.41495168 50.80845876 41.35698824 19.02608903 51.76013532
## [926] 74.93280147 43.58892529 41.03754515 55.26692550 68.29835496
## [931] 31.86174731 42.85936547 46.88183656 53.70377155 59.73406652
## [936] 53.65197405 40.15287690 34.16017622 56.46935995 73.98852599
## [941] 79.01129976 46.96571238 33.92870980 29.77669604 58.56958676
## [946] 63.93608215 35.22621085 32.05672496 13.01625103 59.11802093
## [951] 23.05435156 29.24772501 6.78809878 55.98536762 54.54903682
## [956] 43.75796699 51.16131619 49.76324505 35.07008285 24.77961116
## [961] 57.65930709 80.57855004 56.43145843 47.72260702 59.88635718
## [966] 19.68625020 54.05887379 61.11859249 39.54513110 41.17400518
## [971] 21.64784533 58.04587055 35.97880403 45.14451669 47.06758120
## [976] 52.86041099 53.26266483 45.44228317 5.45196701 64.91185846
## [981] 38.35560102 64.30362718 39.21264764 24.07019481 29.29460108
## [986] 42.25899947 40.45780169 59.03464089 23.45521129 63.16055603
## [991] 41.88971183 58.75635008 22.38228675 23.47540919 42.74780989
## [996] 31.36159134 51.84882712 18.60231350 46.81218069 28.05291386
```

and here is the histogram, where we can see it is roughly normal, as expected:

```
hist(betahat_mat[2,])
```



expect that the mean be very close to 45, which is our true beta₂:

```
mean(betahat_mat[2,])
```

```
## [1] 45.35998
```

We also also expect the standard deviation and the standard error to be very close:

```
sqrt(var(betahat_mat[2,]))
```

```
## [1] 18.16771
```

Since we know that the beta estimate follows $N(\text{truebeta}, \text{truesigma}^2(X^tX)^{-1})$, we can get the standard error as such, confirming they are indeed very similar:

```
sqrt((solve(t(X)%*%X) * truesigma**2)[2,2])
```

```
## [1] 17.80204
```

b)

Focusing on the covariance now, we want to find the 3,4 entry of the covariance matrix:

```
(solve(t(X)%*%X) * truesigma**2)[3,4]
```

```
## [1] 36459.56
```

Comparing it to our empirical(generated) result, we notice that it is double what is expected, which just goes to show the randomness of the generated sample:

```
cov(betahat_mat[3,],betahat_mat[4,])
```

```
## [1] 68171.26
```

Question 2

So let's first load in the data, filter by positive revenue, and sample randomly with my student number as seed:

```
ceocomp <- read.csv('~Documents/GitHub/Big-Ahh-R-Repo/Stat331/Stat331 A2/data/raw/ceocomp.csv')
ceo_pos <- subset(ceocomp, ceocomp$PROF > 0)
set.seed(21066922)
my.ceo_pos = ceo_pos[sample(nrow(ceo_pos), 60),]
my.ceo_pos
```

##	COMP	AGE	EDUCATN	BACKGRD	TENURE	EXPER	SALES	VAL	PCNTOWN	PROF
## 50	616	51	2	3	16	9.0	2754	0.3	0.01	237
## 31	1331	56	2	2	33	4.0	18807	4.8	0.03	942
## 67	925	64	1	4	39	30.0	3670	46.5	2.52	118
## 85	1032	63	2	5	35	8.0	2354	2.2	0.12	46
## 6	2342	71	1	1	46	34.0	835	9.6	0.83	99
## 41	1948	55	1	3	23	23.0	1227	7.6	0.55	145
## 25	768	60	2	1	30	8.0	2097	0.7	0.02	322
## 94	854	62	1	5	30	3.0	669	2.6	0.17	81
## 27	307	47	2	1	14	2.0	649	0.2	0.02	65
## 57	466	48	2	3	17	1.0	1539	0.2	0.01	189
## 95	600	60	0	5	42	1.0	1214	85.6	17.66	43
## 68	596	60	2	4	9	2.0	1148	0.3	0.02	136
## 64	558	62	2	4	34	12.0	2359	0.2	0.01	265
## 49	1031	55	1	3	35	1.0	13340	3.6	0.02	1154
## 74	2094	63	1	5	41	8.0	10818	5.9	0.04	1166
## 12	1030	55	2	1	11	11.0	1236	3.3	0.43	22
## 87	1019	56	2	5	30	9.0	3259	49.3	0.68	311
## 89	600	52	2	5	12	2.0	21351	4.2	0.26	101
## 59	597	57	2	4	27	13.0	2686	1.4	0.03	377

##	32	1034	51	1	2	23	3.0	578	7.0	0.83	82
##	47	1652	60	2	3	34	14.0	14949	11.5	0.06	856
##	16	447	50	2	1	5	1.0	1084	0.3	0.03	54
##	1	570	60	1	1	3	3.0	804	0.2	0.21	49
##	39	517	59	1	2	37	21.0	3180	181.0	6.70	136
##	88	1752	63	1	5	14	14.0	5137	534.2	3.99	249
##	71	809	59	1	5	38	0.5	19196	0.4	0.01	505
##	11	514	57	2	1	3	3.0	661	4.1	0.17	79
##	35	1571	53	1	2	31	9.0	16115	4.2	0.09	301
##	63	2244	64	2	4	41	5.0	4451	4.0	0.04	30
##	79	453	64	2	5	40	26.0	872	54.5	3.14	91
##	62	670	64	2	4	15	15.0	1535	1.9	0.12	145
##	72	2027	62	2	5	25	5.0	8379	3.4	0.04	806
##	76	1126	64	1	5	28	14.0	2545	3.4	0.21	63
##	14	576	48	2	1	8	6.0	1199	0.3	0.02	134
##	70	804	55	0	5	15	10.0	657	10.6	1.75	25
##	30	541	51	2	2	17	12.0	862	1.0	0.10	113
##	9	1341	55	2	1	30	7.0	4377	2.1	0.04	390
##	13	329	37	2	1	11	1.0	344	12.9	4.15	29
##	46	635	59	2	3	33	2.0	3963	0.8	0.12	99
##	96	1088	45	2	5	11	11.0	415	2.6	0.17	48
##	17	1144	60	2	1	37	10.0	4594	6.9	0.37	150
##	44	2847	62	1	3	37	10.0	3817	3.6	0.05	584
##	80	643	65	1	5	29	23.0	1681	8.9	0.89	44
##	48	467	62	2	3	13	8.0	2549	0.3	0.01	306
##	61	893	60	2	4	36	15.0	4444	2.0	0.04	417
##	52	1377	56	1	3	32	3.0	2434	5.3	0.18	273
##	84	493	59	1	5	34	4.0	1556	0.2	0.01	169
##	38	1622	57	1	2	29	5.0	9886	9.7	0.19	347
##	53	621	60	2	3	36	3.0	1552	0.9	0.03	210
##	20	688	54	2	1	28	28.0	3277	1689.0	34.04	317
##	36	1554	61	1	2	41	17.0	1435	30.6	2.23	82
##	37	427	54	1	2	31	15.0	2366	71.4	10.09	55
##	75	1833	63	2	5	36	7.0	17037	6.3	0.04	709
##	91	1630	60	1	5	36	20.0	8331	132.5	8.21	82
##	23	476	50	1	1	20	0.5	3148	1.3	0.05	260
##	45	897	56	2	3	2	5.0	2689	1.0	0.07	81
##	33	1461	63	1	2	23	8.0	5673	3.6	0.05	201
##	69	462	61	2	4	25	11.0	1314	16.8	1.03	27
##	18	425	74	1	1	3	3.0	475	0.1	0.07	6
##	19	1671	63	2	1	17	12.0	1967	5.5	0.20	143

a)

Let's run the model:

```
comp <- my.ceo_pos$COMP
x1 <- my.ceo_pos$AGE
x2 <- my.ceo_pos$EDUCATN
x3 <- as.factor(my.ceo_pos$BACKGRD)
x4 <- my.ceo_pos$TENURE
x5 <- my.ceo_pos$EXPER
x6 <- my.ceo_pos$SALES
x7 <- my.ceo_pos$VAL
x8 <- my.ceo_pos$PCNTOWN
```

```

x9 <- my.ceo_pos$PROF
fullmodel <- lm(comp ~ x1+x2+x3+x4+x5+x6+x7+x8+x9)
summary(fullmodel)

##
## Call:
## lm(formula = comp ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -770.55 -243.91  -58.34   244.04 1377.97
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   47.44104   813.18192    0.058  0.9537
## x1             12.97890    13.09608    0.991  0.3267
## x2            -179.31590   142.78847   -1.256  0.2154
## x32            -56.74491    245.13235   -0.231  0.8179
## x33            -2.10798    213.67373   -0.010  0.9922
## x34           -289.68693    238.66829   -1.214  0.2309
## x35           -106.44351    205.82659   -0.517  0.6075
## x4              11.83513     7.80118    1.517  0.1359
## x5              16.16317    10.72494    1.507  0.1385
## x6               0.01224     0.01796    0.682  0.4989
## x7               0.82042     0.66549    1.233  0.2238
## x8            -58.36783    29.68720   -1.966  0.0552
## x9               0.49202     0.36009    1.366  0.1783
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 500.2 on 47 degrees of freedom
## Multiple R-squared:  0.4259, Adjusted R-squared:  0.2793
## F-statistic: 2.905 on 12 and 47 DF,  p-value: 0.004416

```

b)

Let's run the hypothesis test for $\hat{\beta} = \vec{0}$

```

r2_full <- summary(fullmodel)$r.squared
msreg <- r2_full/12
msres <- (1-r2_full)/47
1 - pf(msreg/msres,12,47)

```

```
## [1] 0.004415791
```

So with a significance level of 0.05, we can conclude with statistical significance that there is at least one explanatory variate that is non-zero, hence contributes to explaining the compensation. ### c)

```

summary(fullmodel)

##
## Call:
## lm(formula = comp ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max

```

```
## -770.55 -243.91 -58.34 244.04 1377.97
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  47.44104   813.18192    0.058  0.9537
## x1           12.97890    13.09608    0.991  0.3267
## x2          -179.31590   142.78847   -1.256  0.2154
## x32          -56.74491    245.13235   -0.231  0.8179
## x33          -2.10798    213.67373   -0.010  0.9922
## x34         -289.68693    238.66829   -1.214  0.2309
## x35         -106.44351    205.82659   -0.517  0.6075
## x4           11.83513     7.80118    1.517  0.1359
## x5           16.16317    10.72494    1.507  0.1385
## x6            0.01224     0.01796    0.682  0.4989
## x7            0.82042     0.66549    1.233  0.2238
## x8          -58.36783    29.68720   -1.966  0.0552 .
## x9            0.49202     0.36009    1.366  0.1783
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 500.2 on 47 degrees of freedom
## Multiple R-squared:  0.4259, Adjusted R-squared:  0.2793
## F-statistic: 2.905 on 12 and 47 DF, p-value: 0.004416
```

- i) with beta hat x34 being -289, we say WITHOUT statistical significance(because the p value is 0.23>0.05) that lawyers tend to make \$289 000 less than banking background, after factoring the rest of the explanatory variates.
- ii) ceos with sales/marketing background also makes less, at -\$56 000, than ones with banking background, accounting for all other explanatory factors. But we're even LESS certain of this estimate, with the p-value at 0.81.

Let's do an additional sum of squares test for the model without industry background as a factor, compared to the full model:

```
redmodel <- lm(comp ~ x1+x2+x4+x5+x6+x7+x8+x9)
r2_red <- summary(redmodel)$r.squared

f_num <- (r2_full - r2_red)/4
f_den <- (1 - r2_full)/47
f_stat <- f_num/f_den
1 - pf(f_stat,4,47)
```

```
## [1] 0.7764592
```

Comparing it to the anova test, we find that the test comes back very similar:

```
anova(redmodel,fullmodel)
```

```
## Analysis of Variance Table
##
## Model 1: comp ~ x1 + x2 + x4 + x5 + x6 + x7 + x8 + x9
## Model 2: comp ~ x1 + x2 + x3 + x4 + x5 + x6 + x7 + x8 + x9
##   Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      51 12201235
## 2      47 11757365  4    443870 0.4436 0.7765
```

Which means that after accounting for the other explanatory variates, we can't conclude the CEO's background

has much effect on the comp. # d) now let's focus on the reduced model:

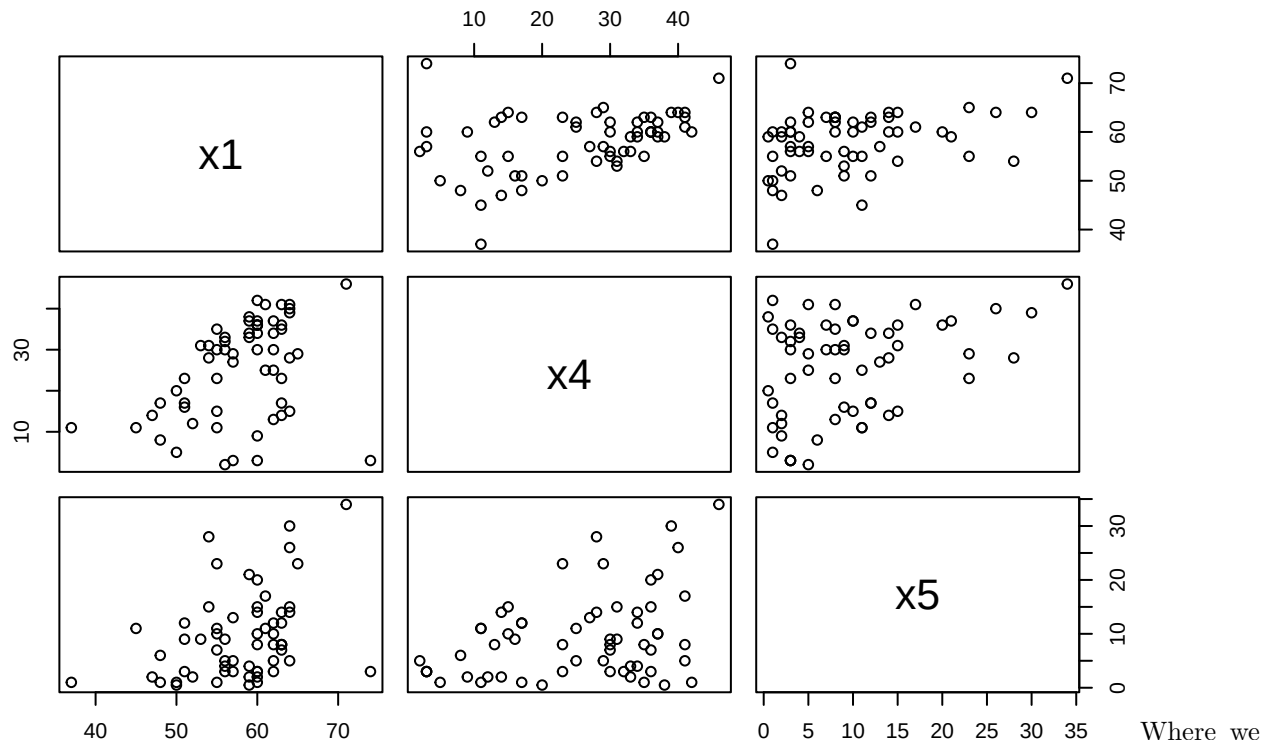
```
summary(redmodel)
```

```
##
## Call:
## lm(formula = comp ~ x1 + x2 + x4 + x5 + x6 + x7 + x8 + x9)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -717.60 -285.72  -39.28   286.79 1311.98
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   277.33972   730.43905    0.380   0.7058
## x1              9.17031    12.13919    0.755   0.4535
## x2            -212.08401    125.88116   -1.685   0.0981 .
## x4             10.44381     7.40399    1.411   0.1644
## x5             15.04509    10.37970    1.449   0.1533
## x6              0.01059     0.01619    0.654   0.5158
## x7              0.88585     0.64852    1.366   0.1779
## x8            -59.34763    28.94672   -2.050   0.0455 *
## x9              0.55425     0.33688    1.645   0.1061
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 489.1 on 51 degrees of freedom
## Multiple R-squared:  0.4042, Adjusted R-squared:  0.3108
## F-statistic: 4.325 on 8 and 51 DF,  p-value: 0.0004988
```

Where we notice that the relation between age, tenure, and experience may be problematic because they are all correlated, as a long EXPER certainly implies longer TENURE and AGE.

Let's see the pairs chart:

```
pairs(cbind(x1,x4,x5))
```



see a slight linear relation between the 3 variables in question. So I'll say that multicollinearity may be an issue. Let's test with the VIF for age:

```
agemodel <- lm(x1~x2+x4+x5+x6+x7+x8+x9)
r2 <- summary(agemodel)$r.squared
VIF <- 1/(1-r2)
VIF
```

```
## [1] 1.419673
```

Which honestly is not that bad, so I retract my statement from before, as the VIF is fairly below 10. In case the assignment was looking for ONLY the 3 variables, here is that VIF:

```
ageredmodel <- lm(x1~x4+x5)
r2 <- summary(ageredmodel)$r.squared
VIF <- 1/(1-r2)
VIF
```

```
## [1] 1.259891
```

Which it seems like it doesn't make much of a difference. ### e) Let's predict a new CEO's comp:

```
new_x <- data.frame(x1=48,x2=1,x4=14,x5=3,x6=3100,x7=2.1,x8=1.1,x9=172)
predict(redmodel,new_x,interval='prediction',level=.95)
```

```
##          fit          lwr          upr
## 1 761.5257 -268.8854 1791.937
```

We can say that the comp of 1.5 million is fairly reasonable, if not a little high. 1500 is still within the 95% prediction interval, so I'd say it is reasonable consistent with the other CEOs' comps.